A certificate-based audit of the fundamental-group and framing computations in Wuebben's proposed exotic $S^2 \times S^2$ construction

John Clyde
VentiMath
verification@ventimath.org

August 2026

UNRELEASED ARTIFACT DRAFT. The v1.5.8 artifact named below is not yet public; do not circulate this PDF until that release exists and this notice is removed.

Abstract

We audit the theorem-critical computations in Wuebben's proposed exotic $S^2 \times S^2$ construction. Four explicit, hash-identified finite presentations associated with his fixed surgery parametrization are proved trivial by derivation certificates accepted by separate Python and Ruby checkers. This finite statement concerns only words and relations. To connect it to geometry, we independently reconstruct the marked genus-two bundle, surgery tori, common whiskers, meridians, and product-framed longitudes, and compare the resulting dictionary with the paper. We expose that dictionary and a worked peripheral extraction here. Exact-rational programs audit the framing calculation that identifies the product and Lagrangian longitudes; its remaining inputs are stated differential- and PL-topological results. Conditional on this geometric identification and twenty-five named external results, a replayed dependency chain recovers the paper's three manifold conclusions. We also record an unresolved contrary computation reported by Wuebben: Lidman and Piccirillo reportedly obtain $\pi_1(V) \neq 1$, but no presentation or nontriviality witness is public, and their paper does not fix the parametrization needed to determine whether the same manifold was computed. Thus the finite presentation theorem is mechanically checkable, while the model-to-manifold identification remains the explicit human trust boundary. The theorems are Wuebben's; this note claims only the audit.

2020 Mathematics Subject Classification. 57R55 (primary); 57K40, 57M05 (secondary).

1 Introduction

For the construction of [1], the theorem-critical step is the fundamental-group computation. The manifolds there rest on a specified Lidman–Piccirillo piece V, obtained by two Luttinger surgeries [8, 3] on Lagrangian tori in a symplectic non-product genus-2 surface bundle R over a once-punctured torus, following the framework of [2]. Write σ for the free orientation-reversing involution of $\partial V = S^3_0(Q)$, Q the square knot, $Z = V \cup_\sigma V$ for the symplectic double, $W =$

V/σ for the quotient, and $Z'' = V \cup_{f\circ\sigma} V$ for the Lidman–Piccirillo regluing. The paper's three theorems are: (A) Z is homeomorphic but not diffeomorphic to $S^2 \times S^2$; (B) W is homeomorphic to Kawauchi's manifold B, and (B, W) is a homeomorphic, non-diffeomorphic pair distinguished by the sliceness of the figure-eight knot; (C) Z'' is a closed simply connected 4-manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. All three rest on the simple connectivity of V. Wuebben fixes one geometric manifold: his $(m, n) = (0, 0)$ member, using two geometric $+1$ Luttinger surgeries and the y_1-based Ax half-drift on T_α. The exponents ϵ_A, ϵ_B in his relation sheet are orientation and presentation conventions, not four geometric parametrizations; he allows either sign for each. We certify all four convention specializations, so in particular the relator sheet for the fixed V is included. The other three specializations are convention-robustness checks, not claims about three further manifolds. Wuebben's own 4,096-system grid varies, besides these two exponents, the three meridian signs of his transport corrections, the left/right placement of those corrections, the choice of conjugating arc for the Bs correction, and the arc, sign, and placement of the T_β push-off. In the rebuilt model none of these is a free choice: the corrections, basings, and push-off are computed as literal certified loops (Section 3), so only ϵ_A, ϵ_B remain as conventions, and the four certificates cover his grid at the fixed member.

This note reports an independent implementation of that step, a machine audit of the framing lemma that connects the group theory to the symplectic geometry, and an explicit proof chain from simple connectivity to the three theorems. Nothing in the implementation reuses the paper's coordinates, code, or intermediate data: the geometry was rebuilt from the paper's stated marked-fiber data alone, and the paper's displayed relations, sign tables, and Table 1 filling words were then *compared against* the rebuild rather than assumed. Every compared item agreed, and the independent raw-paper extractor found no paper-to-model dictionary discrepancy. The downstream deductions are Wuebben's, and this note reproves none of the classification, symplectic, or Floer-theoretic theorems they invoke; what it adds is that each deduction is written out, each theorem invoked is stated with its hypotheses, and each hypothesis is discharged by a named certificate or computation.

For the notation below, "$n = 0$" means Wuebben's Ax half-drift label on T_α; the adjacent "$n = 1$" control replaces Ax by Ar^{-1}. The letters M, N denote based meridians of T_α, T_β, while A, B are the two base generators.

The result has three deliberately separate layers.

Theorem 1 (finite-certificate statement)**.** *Let* `S` *be the byte string*

`verification/luttinger/sealed_transport/r_presentations.json`,

whose SHA-256 digest is

`1e5fc8cb6872a25fa45a227c1e1c207b288c861a3998fe3aae8722ea510c852c`.

Read a positive integer j in `S` *as the generator g_j and a negative integer $-j$ as g_j^{-1}. The top-level fields* `ngens=3` *and* `relators` *define a 3-generator, 78-relator presentation Q. Take the four entries of the top-level array* `paper_fillings` *whose* `half_drift` *field is* `"n0_y1"`*; these are entries* 0–3*, and their (*`sign_a`*,* `sign_b`*) fields run over* $(+1, +1)$*,* $(+1, -1)$*,* $(-1, +1)$*,* $(-1, -1)$ *in that order. Append the two word lists of such an entry, of lengths* 94 *and* 132*, to the* 78 *relators of Q, and call the resulting 3-generator, 80-relator presentation $P_{\epsilon_A,\epsilon_B}$ for $\epsilon_A =$* `sign_a` *and $\epsilon_B =$* `sign_b`*. Every one of these four explicitly defined presentations presents the trivial group.*

The proof consists of four derivation certificates in release `v1.5.8`. Both the Python and Ruby checkers accept them, and two standalone checkers replay the frozen 99,860-step transport that produced `S`. No geometric interpretation of `S` is part of this statement.

Lemma 2 (certificate soundness)**.** *A certificate is a finite sequence of records, each an equation between words in the generators and their inverses, of four kinds: an input relator of the hash-bound presentation; an inverse axiom $gg^{-1} = 1$ of the free group; an overlap, the critical pair of two earlier records, in which each branch is reduced by a recorded trace of rewrites by earlier records; or a change, an earlier record whose two sides are each reduced by such a trace. An identity root for a generator g is an index into the sequence at which the record is $g \to 1$. The checker reads the generator count from the certificate, requires it to equal the count in the hash-bound input presentation, and then requires an identity root for each numbered generator and its inverse. A checker accepts only if every record is derived as stated from records preceding it. Every accepted record is then an equality holding in the group presented by the input relators, so acceptance implies that this group is trivial. Neither the confluence of the completed rewriting system nor the correctness of the program that produced it enters the argument.*

Proposition 3 (model-to-manifold identification)**.** *For every coherent choice of the paper's orientation conventions, the corresponding specialization of Theorem 1 presents $\pi_1(V)$ for the single specified manifold. Thus the four labels express convention choices for one target, not four target manifolds: the simplicial model is identified with the paper's marked bundle by a constructed fiberwise homeomorphism (Section 2); its peripheral loops are based meridians and product-framed longitudes of the surgery tori (Section 3); and the product framing agrees with the Lagrangian framing of Luttinger surgery by the paper's Lemma 8.2, audited by the exact programs and geometric arguments described in Section 3.*

Theorem 4 (downstream implication, relative to named inputs)**.** *For the specified piece V, relative to the twenty-five external theorems listed in Appendix A, the certificates of Theorem 1, the geometric identification of Proposition 3, and the dependency certificate of Appendix A: (A) Z is homeomorphic but not diffeomorphic to $S^2 \times S^2$; (B) W is homeomorphic to B, and the pair (B, W) is homeomorphic but not diffeomorphic, distinguished by the smooth sliceness of the figure-eight knot; (C) Z'' is a closed simply connected 4-manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. Every hypothesis of every external theorem is discharged by a named certificate, computation, or earlier step of the chain, and two independently implemented checkers replay the finite dependency bookkeeping and verify that each of (A), (B), (C) depends on Theorem 1.*

Theorem 1 is a finite statement relative only to the certificate specification and the correctness of the two small checker implementations. Proposition 3 is where the construction-specific mathematical risk lies: it identifies those strings with the intended complement and fillings. Theorem 4 is relative in a different way, since it invokes deep results from the literature. Table 1 records these different evidentiary levels rather than grouping all of them under the word "certified."

The reported contrary computation

Wuebben's public repository has stated, from its first visible commit and continuously since, that Lidman and Piccirillo report a computation of $\pi_1(V)$ that disagrees with his and that the

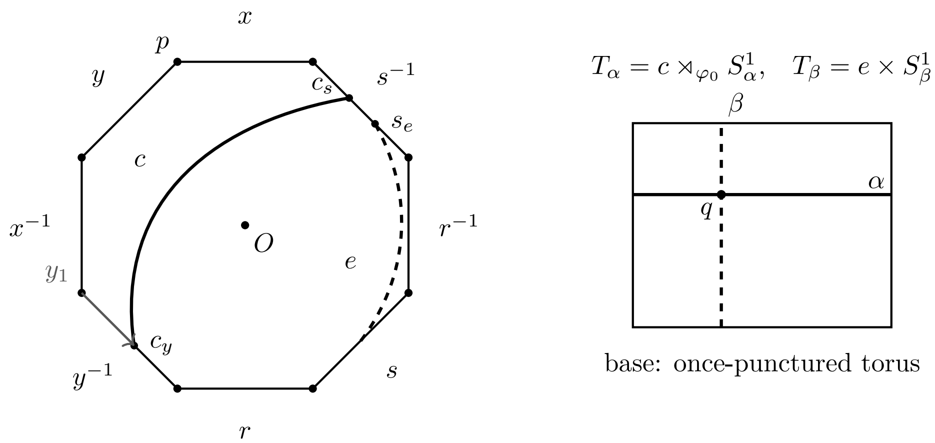
Claim	Public artifact	What replay establishes	Remaining human input
$P_\epsilon = 1$	derivation DAG	every retained equation follows from earlier equations; every generator equals 1	checker implementation conforms to Lemma 2
raw to reduced presentation	frozen Tietze trace	each recorded elimination is valid and all 89 tracked words are transported	the raw presentation describes the geometric complement
peripheral pairs	simplicial paths and normal blocks	paths, common whiskers, orientations, local flatness, and normal-circle fibers	the marked PL model represents the paper's bundle and tori
framing agreement	exact-rational identities and finite hypotheses	all displayed coordinate and exterior-algebra calculations	relative Moser, covering lift, and Weinstein-neighborhood arguments
three manifold conclusions	dependency ledger	hashes, finite calculations, hypotheses recorded, and acyclic dependency	the cited theorems and the semantic correctness of their applications

Table 1: Evidence levels. A replayed finite derivation, a checked combinatorial construction, and a dependency audit are intentionally not treated as the same kind of evidence.

disagreement is unresolved. It is recorded in the README at the commit this verification pins, where the reported computation is said to disagree, and is restated in the current README in the sharper form $\pi_1(V) \neq 1$. It is disclosed here because it concerns the theorem-critical group.

Lidman and Piccirillo's published paper [2] does not assert $\pi_1(V) \neq 1$; it proves $H_1(V) = 0$ and that $\pi_1(V)$ is normally generated by $\pi_1(F)$, which is all their arguments require. The contrary computation is a correspondence report relayed by Wuebben, and we have located no public relation sheet, generator dictionary, or nontriviality witness for it — not in either repository's history, and not in the immutable `v1` source of [2]. Moreover [2] explicitly declines to fix the surgery parametrization, recording in a footnote that because the surgery tori were parametrized somewhat arbitrarily, V is not well defined as such, and that any parametrization serves their claims. Wuebben's theorem, and this verification, concern one fixed member of that family. Whether the reported computation concerns the same member cannot be determined from public materials.

Consequently the two reported outcomes are not known to be about the same manifold, and are equally not known to be about different ones. We claim neither. Theorem 1 is unaffected in either case: it is a statement about explicit finite presentations, and its certificates replay. What a contrary computation of the same fixed member would bear on is Proposition 3, the identification of those presentations with the paper's surgeries — exactly where this note already locates the residual risk. A relation-by-relation comparison is the way to settle it, and that requires the other presentation.



(a) the marked fiber F (b) the two surgery tori

Figure 1: The marked data reconstructed by the audit. Panel (a) is the symmetric octagon model of the genus-two fiber, with boundary word $xyx^{-1}y^{-1}rsr^{-1}s^{-1}$; all eight corners represent the basepoint p, while O is the center. Half-turn carries opposite edges to one another and induces $x \leftrightarrow r$, $y \leftrightarrow s$. The displayed arcs record the curve c, the five-chain curve e, and the initial y_1 whisker; the corresponding s_2 whisker is stored in the simplicial model. Panel (b) is a schematic of the two curve subbundles over the base loops. The geometric placements are checked as simplicial path and incidence assertions; the drawing is explanatory rather than an additional input to the computation.

2 The independent model and the interpretation dictionary

The genus-2 fiber is realized as a regular neighborhood of the paper's five-chain of curves — five plumbed annular bands and two cone discs — with the hyperelliptic-type involution φ_0 realized simplicially; its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R is assembled from simplicial mapping cylinders over the once-punctured torus base; the two surgery tori T_α, T_β arise as induced subcomplexes, and the assembled 4-complex passes manifold checks down to the vertex-link level.

The dictionary used by Proposition 3 is displayed in Table 2; it is not reconstructed from the fact that the filled groups happen to collapse. Here "literal" means a stored edge path in the triangulation, with its basepoint and orientation retained.

The identification with the paper's marked bundle is by literal based monodromy action: the based mapping classes of both monodromies are certified as equalities of vertex paths, transported by proof-producing Tietze chains whose recorded eliminations are replayed in-process by a verifier separate from the generator (these chains are not sealed certificates), and completed by an explicit marked mapping-cylinder clutching over the base graph. The identification is thus a constructed fiberwise homeomorphism into the paper's bundle rather than an appeal to the classification of surface bundles. Everything the surgery argument needs is transported through that map into the paper's already-smooth manifold, where the smooth and symplectic arguments run; no smoothing of the triangulation is ever chosen.

More explicitly, the ordered five-chain fills the surface with two disk complements containing

Paper datum	Marked word or action	Simplicial realization	Check
F, p, O	genus two; two fixed marks	86-vertex closed surface; two fixed vertices	surface and link tests
a, b, c, d, e	$a \sim x$, $b \sim y$, $c \sim xr$, $d \sim s$, $e \sim r$	ordered filling five-chain; oriented c at V_2 is $(rx)^{-1}$	independent ribbon reconstruction
φ_0	$x \leftrightarrow r$, $y \leftrightarrow s$	simplicial half-turn fixing p, O	based path equalities
$\psi_0 = T_a T_b$	$x \mapsto y^{-1}$, $y \mapsto yx$, $r \mapsto r$, $s \mapsto s$	ordered flip stack, T_b first	based monodromy replay
T_α	$c \rtimes_{\varphi_0} S^1_\alpha$	induced torus on the c-stack	local flatness and subbundle check
T_β	$e \times S^1_\beta$	induced product torus on the e-stack	local flatness and product check
A, B	$[\bar{\alpha}]^{-1}, [\bar{\beta}]^{-1}$	literal based loops from q	path export
M, N	meridians based along y_1, s_2	oriented dual normal circles with the same literal whiskers	all 592 normal fibers checked
λ_α	Ax	`lb_a_y1`	explicit construction; see the caveat below
λ_β	$(r^{-1}Mr)B$ in the displayed orientation	`lb_b_s2`, exported without substitution	punctured sweep and independent extraction

Table 2: Paper-to-model dictionary for the fixed $n = 0$ member. Reversing an allowed orientation reverses the corresponding displayed element and is accounted for by (ϵ_A, ϵ_B).

p and O. The independently reconstructed ribbon has intersection matrix

$$i(a,b) = i(b,c) = i(c,d) = i(d,e) = 1, \qquad i(u,v) = 0 \quad \text{otherwise},$$

and the chain-reversing involution has precisely those two fixed marks. Wuebben's Lemma 7.1 therefore identifies it equivariantly with the marked fiber in the paper. On the two-loop spine of the base, the literal based actions are

$$\varphi_0 : (x, y, r, s) \longmapsto (r, s, x, y), \qquad \psi_0 : (x, y, r, s) \longmapsto (y^{-1}, yx, r, s).$$

The alpha identification is equivariant near c; the beta twist annuli are supported away from e. The mapping-cylinder clutching can consequently be chosen relative to both curve subbundles, giving a marked fiberwise homeomorphism $H : |K| \to R_{\text{top}}$ that carries the two induced tori, their collars, and the chosen basepoint to the paper's data.

Because the machine certifies properties of a model, the model itself is the residual risk. Five structurally different implementations reconstruct it independently, arranged so that each class of translation error — subdivision artifacts, monodromy transcription, ordering, sign, and orientation conventions — would be caught by at least one of them. What redundancy cannot exclude, a shared misreading of the paper's conventions, is addressed by a bound interpretation dictionary: every convention read from the paper carries the certified computation that would fail under a rival reading. Wuebben's committed scripts enter at exactly one point, parsed and never executed: after the dictionary was frozen, selected entries were compared diagnostically against them, and no verified conclusion consumes script-derived data. All selected entries agreed. The remaining transcription-level entries are audited by a quarantined extractor against the hash-pinned immutable Lidman–Piccirillo v1 arXiv source [2], in both its prose and its vector-figure layers; the one interpretation remaining at that boundary is the ordinary convention that dashed arcs in a handle picture depict hidden projection rather than extra crossings. That extractor distinguished the nearest rival twist-order and twist-sign readings and found no discrepancy with the independently built model. The full inventory of implementations and the dictionary are in the repository.

3 Peripheral data and the framing identification

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the paper's own basing whiskers. The paper's sign-correction package (the corrected relations of its §8.3 and the push-off basing correction of its §8.5) is reproduced independently, and a combinatorial sweep of the relevant transport square resolves the basing double-count question in the direction the paper's corrected convention requires. Both surgery tori are certified locally flat at every simplex link, and every extracted dual meridian is checked to be a literal normal-circle fiber of an explicit normal block model, with no regular-neighborhood theorem invoked. The closed section $\widehat{\Gamma}$ of the doubled bundle — the union of the two copies of the section through a fixed point of φ_0 — is extracted as a closed orientable genus-2 surface, and its self-intersection $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$ is certified with an explicit normal push-off; this number is used directly in Appendix A.

The relation sheet and the two fillings

To make Proposition 3 inspectable without opening the repository, fix the orientation sheet used by the triangulation and put $\delta = r^{-1}$. Before filling, the paper-coordinate presentation read from the literal paths has the surface relation, the following eight transport relations, and the drilled-fiber relation:

$$\begin{aligned} &AxA^{-1} = r, \quad AyA^{-1} = s, \quad ArA^{-1} = x, \quad AsA^{-1} = Ny, \\ &BxB^{-1} = y^{-1}, \quad ByB^{-1} = M^{-1}yx, \quad BrB^{-1} = r, \quad BsB^{-1} = (r^{-1}M^{-1}r)s, \\ &[x,y][r,s] = 1, \quad B(s^{-1}r^{-1}yx)B^{-1} = r^{-1}s^{-1}x. \end{aligned}$$

The two commonly based product-framed directions are

$$\lambda_\alpha = Ax, \qquad \lambda_\beta = (r^{-1}Mr)B,$$

and the four convention sheets append exactly

$$M\lambda_\alpha^{\epsilon_A} = 1, \qquad N\lambda_\beta^{\epsilon_B} = 1, \qquad (\epsilon_A, \epsilon_B) \in \{\pm 1\}^2. \tag{1}$$

Changing an allowed orientation inverts the corresponding meridian or displayed longitude and changes the sheet label; it does not define a new geometric surgery. The finite presentations of Theorem 1 are the frozen Tietze images of these literal path presentations, but that geometric assertion is part of Proposition 3, not of the finite theorem.

Worked peripheral extraction

Consider T_α. The curve c meets the oriented y edge once, at c_y. Starting at p, the stored path follows the initial segment y_1 to c_y, makes a normal jog to the frontier of the explicit normal block, and traverses the oriented dual normal circle. Returning along the same whisker gives M. The longitude uses that identical whisker and normal jog, traverses the base direction A, and closes along the selected half of c. The stored words are `geom_M` and `lb_a_y1`; they are what M and λ_α denote throughout, and they are what Theorem 1 literally fills with.

That the two loops use the same whisker is not an inference but a fact one can read off the sealed transport input: there `geom_M` $= y_1\,\mu_\alpha\,y_1^{-1}$ and `lb_a_y1` $= y_1\,\lambda_\alpha^\circ\,y_1^{-1}$ with the *identical* four-edge whisker $y_1 = e_{24046}e_{90328}e_{79763}e_{72645}$, seven- and six-edge circles $\mu_\alpha, \lambda_\alpha^\circ$, and fifteen and fourteen letters in all. So no independent conjugation is hidden in the filling word $M\lambda_\alpha^{\epsilon_A}$.

The identification of the stored word `lb_a_y1` with the paper's symbol Ax has two independent supports. Geometrically it follows from how the path is built — based on the y rail at the unique crossing c_y, whiskered by y_1, jogged to the normal-block frontier, lifting the base generator A, and closed along the selected half of c. Algebraically, the elementary elimination pass over the sealed presentation leaves the 72-letter residual `lb_a_y1`^{-1}Ax, but a proof-producing completion reduces that exact word to the identity in 61 steps. Independent Python and Ruby verifiers replay the complete 2,506-record ancestry cone from the sealed three-generator, 78-relator complement; neither filling relation is present. The frozen source, certificate, compiler, and both verifiers are in `verification/luttinger/alpha_residual/` (Run 68). An earlier run had reported an empty elementary residual over a presentation predating both committed exports. That evidence does not reproduce and is superseded by the sealed certificate just described.

The beta-coordinate word `lb_b_s2`$^{-1}(r^{-1}Mr)B$ likewise leaves a 113-letter residual under elementary elimination. The same proof-producing complement system reduces that exact word to the identity in 82 steps; independent Python and Ruby verifiers replay its 1,540-record ancestry cone. The frozen source, certificate, compiler, and both verifiers are in `verification/luttinger/beta_residual/` (Run 69). Thus both displayed coordinate identities are algebraically certified in the complement, although Theorem 1 remains more direct: it fills with the literal stored boundary longitudes rather than relying on either coordinate substitution.

The beta extraction is the sign-sensitive case. The corresponding path uses the initial segment s_2 to the unique crossing s_e, the base loop B, and the fiber-normal push-off. Its swept basing square crosses T_α once. Puncturing that square contributes the conjugated meridian $r^{-1}Mr$, which is why the paper's word for this direction is $(r^{-1}Mr)B$ and not B. The stored words are `geom_N` and `lb_b_s2`, exported without substitution; they are what N and λ_β denote, and the identification with $(r^{-1}Mr)B$ carries the same caveat as the α case. An independently written extractor recovers both pairs without importing the complement, sweep, or peripheral modules. Local link checks certify both tori at every simplex, and the explicit block-frontier map checks every dual loop as a normal-circle fiber; these are the finite facts behind the geometric words "meridian" and "push-off."

From product framing to Lagrangian framing

Luttinger surgery is performed with respect to the Lagrangian framing; the paper's Lemma 8.2 identifies it with the fibered framing carried by the combinatorial longitudes above. This lemma carries an exact logical load: if the identification failed by j meridians, the certificates of Theorem 1 would remain correct for the product-framed presentations they literally describe, but would no longer identify those presentations with the paper's surgeries. The lemma's displayed Weinstein-chart, seam, and constant-momentum algebra are checked by exact-rational Python programs, and its inline analytic normalizations are rebuilt constructively rather than cited: the relative Moser flow is produced as the explicit monotone inverse of a cumulative coordinate, and the equivariant step defines the Moser field directly on the double cover with certified projection and deck invariance, so neither an ODE existence theorem nor a covering-lift theorem is invoked. Chart independence is proved between the paper's displayed charts by two routes, with the one symplectic move that could introduce meridian components — the closed-1-form shift — excluded by the zero-section condition; [3] (§2.1 and Proposition 2.2) stands as corroboration. These programs verify the displayed identities and the finite hypotheses of the local constructions; they are not a foundational formalization of differential topology.

The local calculation itself is short enough to display. Near T_β, ψ_0 is the identity near e, and relative Moser gives

$$\omega = dt \wedge d\theta_1 + K\, du \wedge d\theta_2;$$

the in-fiber and base-normal push-offs have constant momentum. Near T_α, equivariant Moser normalizes the free collar involution to $(\theta_1, t) \mapsto (\theta_1 + \pi, t)$. On the mapping-torus quotient put

$$\Theta_1 = \theta_1 - \tfrac{1}{2}\theta_2, \quad \Theta_2 = \theta_2, \quad P_1 = t, \quad P_2 = \tfrac{1}{2}t + Ku.$$

These functions respect the seam and satisfy

$$dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2 = dt \wedge d\theta_1 + K\, du \wedge d\theta_2.$$

The fiber push-off has momentum $(t_0, t_0/2)$ and the half-drifting base push-off has momentum $(0, Ku_0)$, both constant. Hence neither acquires a meridian component, which is precisely the equality of framings used in (1). The exact-rational checker verifies the seam, exterior-algebra, primitive, positivity, and constant-momentum identities; the remaining human inputs are the stated relative Moser and Lagrangian-neighborhood arguments.

Proof of Proposition 3. The ordered filling ribbon, its two complementary disks, the involution, and the fixed marks satisfy the hypotheses of Wuebben's Lemma 7.1, so the equivariant marked-fiber identification described in Section 2 exists. Equality of the two literal based monodromies extends it over the two mapping-cylinder handles of the base. The alpha conjugacy is equivariant near c, and the beta twists have support disjoint from e; hence the extension is relative to the two curve subbundles and carries the induced tori and their product collars to T_α and T_β.

The local-flatness and normal-block checks identify the two oriented dual loops as meridians. The worked extractions build their common-whisker product push-offs as the paper's Ax and $(r^{-1}Mr)B$ directions — by construction, subject to the algebraic caveat recorded there — with the punctured beta sweep accounting for the conjugated meridian.

It remains to place the drilled-fiber relation $B(s^{-1}r^{-1}yx)B^{-1} = r^{-1}s^{-1}x$ in the complement rather than in the mapping cylinder. This step is load-bearing, since Wuebben's presentations become tractable only after it is adjoined, and an earlier form of this audit established only the based monodromy identity in the *unpunctured* mapping cylinder, comparing a non-discriminating low-index fingerprint in the complement; neither step showed that the transport homotopy avoids the surgery tori, which is where the relation is used. The distinction is not vacuous, and the checker exhibits why: the transported loop κ_3, of edge length 31, misses c entirely but meets e once, so sweeping it along the *core* of the beta curve would meet T_β in 264 vertices. Wuebben specifies the parallel push-off, not the core. Carried along the push-off, the complete mapping-cylinder stack — 17,664 tetrahedra on 2,966 vertices — has no vertex on T_α and none on T_β; a 2×35 simplicial homotopy grid at p then joins the push-off basepoint loop to the core one and identifies it with the presentation's whiskered generator, again meeting neither torus; and the based monodromy target reduces to $r^{-1}s^{-1}x$ as required. The relation therefore holds in $R \smallsetminus (T_\alpha \cup T_\beta)$. This repairs an omission in the identification argument, not in the certificates of Theorem 1, which are finite objects about explicit presentations. The audit is `luttinger/r3_complement_audit.py`, SHA-256 `76e579c18227…eefb79`.

Finally, the coordinate calculation above, together with the stated relative Moser and Lagrangian neighborhood inputs, identifies the product and Lagrangian framings. Therefore the two normal closures added to the model are exactly

$$\langle\!\langle M\lambda_\alpha^{\epsilon_A}, N\lambda_\beta^{\epsilon_B} \rangle\!\rangle,$$

the paper's two Luttinger filling relations on the fixed geometric member. Changing a common whisker conjugates both elements of a peripheral pair, and changing an allowed orientation merely changes the convention sheet. Thus each of the four sheets attached to Theorem 1 presents $\pi_1(V)$ for the same specified manifold. □

As auxiliary robustness evidence, a cross-and-diagonal sample of 96 meridian-shifted counterfactual fillings, together with four zero-shift controls, all returned the trivial group. Only 22 of those 100 cases have independently replayed derivation certificates; the other 78 retain GAP session verdicts. Because the scan does not discriminate the correct framing from the sampled incorrect ones, it carries no logical load for Proposition 3; the complete inventory and search history are kept in the repository rather than repeated here.

4 Certified triviality of the four convention-specialized groups

For each of the four $n = 0$ entries defined in Theorem 1, KBMAG produced a complete rewriting system in which all three numbered generators reduce to the identity. Completion is only the discovery method: each verdict is exported as a derivation DAG and proved by Lemma 2. The certificate sizes are shown below.

system	signs	retained records
n=0	$(-,-)$	3620
n=0	$(-,+)$	3803
n=0	$(+,-)$	2470
n=0	$(+,+)$	6672

Retained derivation-DAG records over the sealed presentation. Each certificate verifies triviality from a fresh clone.

An earlier four-generator export, four adjacent $n = 1$ controls, calibration examples, deliberate corruptions, and the full framing-scan history are summarized in Appendix C. None carries logical load for the fixed $n = 0$ target.

5 Certificate verification

Knuth–Bendix completion (KBMAG 1.5.9 [25], run under GAP 4.11.1 [26]) is used only as an *untrusted certificate generator*. A patched build of KBMAG records the ancestry of every equation it derives; the patch is distributed as `verification/luttinger/kbmag-proof/kbfns-proof.patch`. An untrusted compiler prunes that history to the dependency cone of the final identity rules and emits a certificate of the form in Lemma 2, binding the input presentation by SHA-256. The certified set is enforced, not assumed: both checkers reject a batch that verifies any case twice, and in full-inventory mode — the documented invocation — require the batch to be exactly the input's eight fillings, one file named by each case slug, over a source of the stated shape (three generators and 78 complement relators for the sealed chain, four and 95 for the earlier export) with two filling relators per case; the manifest generator refuses to hash a certificate directory that is not exactly one file per filling. A successful replay therefore verifies exactly the four $n=0$ sign pairs and the four $n=1$ controls. A small independent checker then re-verifies every input-relator axiom, inverse axiom, critical overlap, rewrite trace, tidying change, and final generator-to-identity rule; it neither imports nor runs KBMAG. A second checker, written independently in Ruby, re-verifies all retained derivation records against the same hash-bound certificates and accepts, importing neither the first checker nor any project group code. All certificates, the exported rewriting systems, and a hash manifest replay from a clean checkout, and the stored rewriting systems reload and re-verify under GAP/KBMAG independently of the certificate path.

The transport from the raw simplicial presentation to the presentation whose fillings are certified is sealed in the same sense. The canonical seeded run serializes the raw complement presentation (99,863 generators, 321,702 relators) together with its 89 tracked peripheral and coordinate words, records the 99,860 elementary eliminations that reduce it to the 3-generator, 78-relator presentation, and freezes that output. Two standalone checkers — one in Python, one in Ruby, importing neither the geometry nor the simplifier — replay the eliminations from those three frozen files alone: a step is accepted only if the eliminated generator occurs exactly once in the named current relator and the recorded replacement is the one that relator implies, input and output are bound by SHA-256, and the renumbered result must equal the frozen presentation relator by relator and tracked word by tracked word. Both reject corrupted-input, omitted-step, altered-substitution, and altered-output controls. The chain from the frozen raw complex to the eight triviality verdicts (39,163 retained derivation records over the sealed presentation) therefore replays end to end from committed files. The design principle throughout is that certificates are checked, not trusted: search and completion tools are untrusted generators, and every computational claim rests on a replayable artifact validated by small independent code.

The 78-relator input has $H_1 = \mathbb{Z}^2$, and each filling pair kills its abelianization; triviality rather than mere perfectness is carried by the derivation certificates. The negative controls and calibration examples in Appendix C show that the pipeline also produces nontrivial groups and that both checkers reject altered records and incomplete inventories. Those controls test implementation behavior; they are not substitutes for the soundness argument above.

A From simple connectivity to the three theorems

The deductions from $\pi_1(V) = 1$ to Theorems A, B, C occupy Part I of [1], whose Theorem 1.2 reduces them to the simple connectivity of V. This section carries those deductions through as an explicit dependency chain. The chain is itself a machine artifact: a script records every item as one of four kinds — an *external theorem* stated with its hypotheses and source; a *certificate* of this repository, bound by SHA-256; a *computed* fact, executed by the script; or a *step*, a deduction from earlier items — and freezes the result as a certificate. A second script, written independently in Ruby, re-derives every computed fact from scratch, checks that every citation resolves and that the dependency graph is acyclic, verifies that each of the three conclusions depends on Theorem 1, and checks every evidence file against its recorded digest. The chain has 25 external theorems, 3 certificates, 15 computed facts, and 16 steps; its prose form is in the repository, and what follows is the mathematics, with the inputs first. Both scripts in this downstream pair were produced by Claude Fable 5; independence here likewise means separate implementations, not cross-model authorship.

External inputs

Elementary algebraic topology enters as: the Seifert–van Kampen theorem for a union along a connected bicollared subspace; the exact sequence $1 \to \pi_1(Z) \to \pi_1(W) \to G \to 1$ of a connected regular covering with deck group G; Poincaré–Lefschetz duality and universal coefficients, with the Euler characteristic as an alternating sum of Betti numbers over any field, multiplicative for bundles with compact fiber and additive along a common boundary; the index formula $\det(\operatorname{Gram} S) = [L : S]^2 \det(\operatorname{Gram} L)$ for a finite-index sublattice, with the unimodularity of the intersection form on H_2/Tors; Wu's formula $\langle w_2, x\rangle = x \cdot x$ and its consequence $\langle w_2(X), [F]\rangle = \chi(F) + F \cdot F \pmod 2$ for an embedded oriented surface; Novikov additivity of the signature; asphericity of bundles with aspherical fiber and base; the adjunction formula $K \cdot S + S \cdot S = 2g - 2$ for a symplectic surface; and the isotopy of any two orientation-preserving smooth embeddings of B^4 in a connected 4-manifold [24], which together with the amphichirality of the figure-eight knot makes smooth sliceness in a closed 4-manifold a diffeomorphism invariant.

The named theorems are the following.

- **Freedman** [9]: closed simply connected topological 4-manifolds are classified up to homeomorphism by the intersection form and the Kirby–Siebenmann invariant; every unimodular form is realized, uniquely if even (with $\mathrm{KS} \equiv \sigma/8$), and by exactly two manifolds if odd, one per value of KS.
- **Hambleton–Kreck** [10, 11, 12], in the form of [13, Theorem 5.1]: "Closed, oriented topological 4-manifolds with finite cyclic fundamental groups are classified up to homeomorphism by $\pi_1(M)$, q_M, the w_2-type, and $\mathrm{KS}(M)$," where q_M is the form on H_2/Tors and the w_2-type is (II) for spin M.
- **Thurston** [16] and the paper's Thurston-type form ([1], proofs of Proposition 1.3 and Lemma 8.2; [2], proof of Theorem 8): a closed surface bundle over a closed surface with homologically essential fiber is symplectic, and for $R \cup_\sigma R$ the form can be chosen positive on the fiber F and on the closed section $\widehat{\Gamma}$, with the surgery tori Lagrangian.

- **Luttinger surgery** [8, 3]: surgery on a Lagrangian torus yields a symplectic manifold, agreeing with the original outside the surgered neighborhood and with unchanged Euler characteristic.

- **Symplectic Kodaira dimension** [15], as summarized in [14, p. 1]: κ is defined from a minimal model and "is independent of the choice of symplectic form ω"; a minimal symplectic 4-manifold has $\kappa = -\infty$ iff it is rational or ruled, and rational and ruled 4-manifolds have $\pi_2 \neq 0$. **Ho–Li** [14, Theorem 1.1]: "The Luttinger surgery preserves the symplectic Kodaira dimension."

- **Symplectic Thom** [17]: an embedded symplectic surface in a closed symplectic 4-manifold is genus-minimizing in its homology class; and the construction ([1], proof of Lemma 4.3, after [20]) of a connected symplectic unbranched k-fold cover of a square-zero symplectic surface inside its tubular neighborhood, representing k times its class.

- **Klug** [21, Theorem 2]: "Let X^4 be a smooth compact connected oriented 4-manifold with ∂X an integer homology sphere. Let F^2 be an orientable characteristic surface with connected boundary that is properly embedded in X. Then $\mathrm{Arf}(F)+\mathrm{Arf}(\partial F) = (\sigma(X)-[F]^2)/8+\mu(\partial X) \pmod 2$." For spin X, characteristic means vanishing in $H_2(X, \partial X; \mathbb{Z}/2)$. **Levine** [22]: $\mathrm{Arf}(K) = 0$ iff $\Delta_K(-1) \equiv \pm 1 \pmod 8$.

- **Slice disks and the 0-trace** ([2], proof of Theorem 1): a smooth slice disk with Seifert-framed normal bundle embeds the simply connected 0-trace, in which a capped Seifert surface is a closed square-zero surface generating H_2; the framing condition is automatic when the relative H_2 is torsion; the figure-eight knot has genus 1.

- **Lidman–Piccirillo's constructions** [2]: σ is a free orientation-reversing bundle involution (Lemma 4); the sections from the fixed points of φ_0 survive into V, their boundary circles are preserved by σ, and $R \cup_\sigma R$ is a closed genus-2 bundle over a closed genus-2 surface on which the surgeries act by Luttinger surgery on four disjoint tori missing F and $\widehat{\Gamma}$ (Lemma 6, proof of Theorem 8); $W = V/\sigma$ is a closed smooth oriented 4-manifold, double covered by Z, and is spin by the hyperbolic-pair argument of Lemma 7, whose hypotheses are the homology of V; the regluing diffeomorphism f of $S^3_0(Q)$ exists, and with $\mathcal{A} \subset S^3_0(Q)$ the framed annulus joining $\partial\Gamma$ to its image under $f \circ \sigma$, the closed surface $\Gamma'' = \Gamma \cup_{\mathcal{A}} \Gamma$ has odd square and meets F once (proof of Theorem 2); and the Floer-theoretic input, Lemmas 9 and 10 with the vanishing of all mixed invariants of $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ (and of its sum with any homology 4-sphere) along both square-zero lines, via splittings over $S^2 \times S^1$ (proof of Theorem 2, using [18, 19]).

- **Kawauchi's manifold** [23, 2]: B is a closed smooth oriented spin 4-manifold with $\pi_1(B) = H_1(B) = \mathbb{Z}/2$ and $b_2(B) = 0$, in which the figure-eight knot is smoothly slice.

The chain

Throughout, F is a fiber of R disjoint from the surgery tori, Γ a section through a fixed point of φ_0, $\widehat{\Gamma} = \Gamma \cup_\sigma \Gamma \subset Z$ and $\Gamma'' = \Gamma \cup_{\mathcal{A}} \Gamma \subset Z''$ its closures, $\mathcal{A}$ the framed annulus above. The finite calculations cited are those of the chain's certificate.

Proposition 5. *$H_1(V) = 0$, $H_3(V) = 0$, $H_2(V) = \mathbb{Z}\langle F\rangle$ with $F \cdot F = 0$, and V is spin. Moreover $\pi_1(Z) = \pi_1(Z'') = 1$.*

Proof. Wuebben's Lemma 4.2 and the two filling relations give $H_1(V) = 0$ independently of simple connectivity; the certificate also implies it. The boundary $S^3_0(Q)$ is connected, so $H^1(V, \partial V) = 0$ and hence $H_3(V) = 0$, and $H_1(V, \partial V) = 0$ so $H_2(V)$ is torsion-free. $\chi(V) = \chi(R) = \chi(F)\chi(\text{base}) = (-2)(-1) = 2$, unchanged by the surgeries, and $b_0 = 1$, $b_1 = b_3 = b_4 = 0$ give $b_2(V) = 1$. A section is a properly embedded surface meeting F once, so $[F]$ is primitive and generates. $F \cdot F = 0$ since a fiber has trivial normal bundle. $H_2(V; \mathbb{Z}/2) = \mathbb{Z}/2\langle F\rangle$, w_2 is detected on it, and $\langle w_2, F\rangle = \chi(F) + F \cdot F = -2 + 0 \equiv 0$. Finally Z and Z'' are unions of two copies of V along the connected bicollared boundary, so by van Kampen each fundamental group is a quotient of $\pi_1(V) * \pi_1(V) = 1$. This is the one downstream step where simple connectivity of V, rather than only of Z, is needed: the regluing twists the amalgam. □

Proposition 6. *$H_2(Z) = \mathbb{Z}\langle F, \widehat{\Gamma}\rangle$ with intersection form the hyperbolic form H; Z is closed, simply connected, spin, with $b_2 = 2$ and signature 0; and Z is homeomorphic to $S^2 \times S^2$.*

Proof. $\chi(Z) = 2\chi(V) - \chi(S^3_0(Q)) = 4$ and $\pi_1(Z) = 1$ give $b_2(Z) = 2$ with $H_2(Z)$ torsion-free. $F \cdot F = 0$, $F \cdot \widehat{\Gamma} = 1$ (a fiber meets a section once), and $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$ by the certified self-intersection computation of Section 3. The Gram determinant is -1 and the form is unimodular, so the index formula makes $(F, \widehat{\Gamma})$ a basis and the form is H. H is even, so $w_2(Z) = 0$ by Wu's formula. For even forms $\mathrm{KS} \equiv \sigma/8 = 0$ is determined, so Freedman's classification gives $Z \cong S^2 \times S^2$. Note that spinness of Z is obtained here from the certified $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$; the paper's route through the spin quotient W is not needed for this step. □

Proposition 7. *Z is a closed symplectic 4-manifold in which F and $\widehat{\Gamma}$ are symplectic surfaces of genus 2 and square 0, and Z is not diffeomorphic to $S^2 \times S^2$.*

Proof. $R \cup_\sigma R$ is a closed surface bundle whose fiber is homologically essential ($F \cdot \widehat{\Gamma} = 1$), so it carries the Thurston-type form, positive on F and $\widehat{\Gamma}$. The surgery tori are Lagrangian for it and the surgeries are the certified Luttinger surgeries (Proposition 3), and Luttinger surgery preserves the form away from the surgered neighborhoods, which miss F and $\widehat{\Gamma}$. For the non-diffeomorphism: $R \cup_\sigma R$ is aspherical, hence minimal (an exceptional sphere would be null-homotopic, hence of square 0, not -1) and neither rational nor ruled (those have $\pi_2 \neq 0$), so $\kappa(R \cup_\sigma R) \neq -\infty$; Luttinger surgery preserves κ, so $\kappa(Z) \neq -\infty$. If $\psi\colon Z \to S^2 \times S^2$ were a diffeomorphism, composing with a reflection of one factor if necessary makes it orientation-preserving; ψ^* of a product form is then a symplectic form on Z inducing its orientation with $\kappa = \kappa(S^2 \times S^2) = -\infty$, contradicting the independence of κ from the form. □

Propositions 6 and 7 are Theorem A.

Proposition 8. *W is a closed oriented smooth spin 4-manifold with $\pi_1(W) = \mathbb{Z}/2$, $b_2(W) = 0$, signature 0, and $\mathrm{KS}(W) = 0$; and W is homeomorphic to B.*

Proof. $Z \to W$ is a connected 2-fold covering, so $|\pi_1(W)| = 2$. $\chi(W) = \chi(Z)/2 = 2$ and $b_1(W) = 0$ give $b_2(W) = 0$, hence signature 0. W is spin by Lidman–Piccirillo's Lemma 7, whose hypotheses Proposition 5 supplies, and $\mathrm{KS}(W) = 0$ since W is smooth. W and B are then closed oriented topological 4-manifolds with the same Hambleton–Kreck invariants: $\pi_1 = \mathbb{Z}/2$, the zero form on $H_2/\mathrm{Tors} = 0$, w_2-type (II), $\mathrm{KS} = 0$. □

Proposition 9. *No nonzero square-zero class in $H_2(Z;\mathbb{Z})$ is represented by a smoothly embedded torus.*

Proof. For $k \geq 1$ the connected symplectic k-fold covers of F and of $\widehat{\Gamma}$ represent kF and $k\widehat{\Gamma}$ and have genus $k+1$, since $\chi = -2k$; by the symplectic Thom theorem these genera are minimal, and reversing orientation covers $k < 0$. Since $(aF + b\widehat{\Gamma})^2 = 2ab$, every nonzero square-zero class is kF or $k\widehat{\Gamma}$ with $k \neq 0$, of minimal genus $|k| + 1 \geq 2$. $\square$

Proposition 10. *The figure-eight knot is not smoothly slice in W.*

Proof. Let D be a smooth slice disk in $W^\circ = W \setminus B^4$. The group $H_2(W^\circ, \partial; \mathbb{Z}) \cong H^2(W) \cong \mathbb{Z}/2$ is torsion, so the relative self-intersection of D is 0 and its framing is the Seifert framing. If $[D, \partial D] = 0$ in $H_2(W^\circ, \partial; \mathbb{Z}/2)$, then D is characteristic in the spin manifold W°, and Klug's theorem with $\mathrm{Arf}(D) = 0$, $\sigma(W^\circ) = 0$, $[D]^2 = 0$, $\mu(S^3) = 0$ forces $\mathrm{Arf}(4_1) = 0$; but $\Delta_{4_1}(t) = -t^2 + 3t - 1$ gives $\Delta_{4_1}(-1) = -5 \equiv 3 \pmod 8$, so $\mathrm{Arf}(4_1) = 1$. Otherwise the 0-trace $X_0(4_1)$ embeds in W and the torus T obtained by capping a Seifert surface with D has square 0 and $[T] \neq 0$ in $H_2(W;\mathbb{Z}/2)$. Since $X_0(4_1)$ is simply connected it lifts to Z, and T lifts to a torus $\tilde{T}$ with $p_*[\tilde{T}] = [T] \neq 0$ in $\mathbb{Z}/2$ coefficients; so $[\tilde{T}] \neq 0$ in $H_2(Z;\mathbb{Z}/2) = H_2(Z;\mathbb{Z}) \otimes \mathbb{Z}/2$ (as $H_1(Z) = 0$), and a class with nonzero mod-2 reduction is nonzero in the torsion-free $H_2(Z;\mathbb{Z})$; and $\tilde{T} \cdot \tilde{T} = T \cdot T = 0$ because p is a local diffeomorphism. This contradicts Proposition 9. $\square$

Proposition 8, Proposition 10, the sliceness of the figure-eight knot in B, and the diffeomorphism invariance of sliceness give Theorem B.

Proposition 11. *Z'' is closed, simply connected, smooth, with $b_2 = 2$, signature 0, and odd intersection form $\langle 1 \rangle \oplus \langle -1 \rangle$; hence Z'' is homeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.*

Proof. $\pi_1(Z'') = 1$ and $\chi(Z'') = 4$ give $b_2 = 2$ with H_2 free. Both Z and Z'' decompose into the same two oriented copies of V along their common boundary, so Novikov additivity gives $\sigma(Z'') = \sigma(Z) = 0$, independently of the regluing map. $F \cdot F = 0$, $F \cdot \Gamma'' = 1$, and $\Gamma'' \cdot \Gamma'' = 2\ell + 1$ is odd; the Gram determinant is -1, so (F, Γ'') is a basis, and $E = \Gamma'' - \ell F$, $D = F - E$ satisfy $E \cdot E = 1$, $D \cdot D = -1$, $E \cdot D = 0$ for every $\ell \in \mathbb{Z}$: the form is $\langle 1 \rangle \oplus \langle -1 \rangle$. Of Freedman's two manifolds with this form the one with $\mathrm{KS} = 0$ — Z'' is smooth — is $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. $\square$

Proposition 12. *Z'' is not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.*

Proof. Z is symplectic with $|K_Z \cdot F| = 2$ by adjunction (F symplectic of genus 2 and square 0) and $H^2(V) = \mathbb{Z}$, so Lidman–Piccirillo's Lemma 9 gives $\Psi_{V,t} \neq 0$ for both spin^c structures with $|\langle c_1(t), F \rangle| = 2$. Z'' is $V \cup V$ along $S^3_0(Q)$ with t restricting to the non-torsion $s_\pm$, $b_2^+(Z'') = 1$, and $\mathrm{span}\{F\}$ a square-zero line, so Lemma 10 gives a nonvanishing mixed invariant $\Phi_{Z'', \mathrm{span}\{F\}, t}$. A diffeomorphism to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ would carry $\mathrm{span}\{F\}$ to one of its two square-zero lines — in $\langle 1 \rangle \oplus \langle -1 \rangle$, $a^2 - b^2 = 0$ forces $a = \pm b$ — along each of which the manifold splits over $S^2 \times S^1$ and every mixed invariant vanishes. $\square$

Propositions 5, 11 and 12 are Theorem C, and Theorem 4 is proved.

Three points where the chain's route differs from the paper's are worth recording. $H_1(V) = 0$ is taken from the certificate rather than from the paper's Lemma 4.2 and the surgery relations, which are needed only for the "forcing" direction of its Proposition 1.3. The intersection form of Z is identified from the certified $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$, so that Z is spin because its form is even; W spin still enters, for the Hambleton–Kreck invariants and for Klug's theorem. And the sliceness dichotomy is taken over $\mathbb{Z}/2$ throughout, which is exactly what makes the null-homologous case Klug's characteristic case and the lifted torus nonzero integrally.

B Trust boundary and data availability

Checker provenance and audit status

The verification was developed by Anthropic Claude models (Fable 5 and Opus 5) and OpenAI Codex (GPT-5 family) under the author's direction, with commit-level provenance and cross-model corrections retained in the public ledger. Mathematical responsibility rests with the human author; model provenance is not evidence of correctness.

Both filled-group checkers were produced as separate Python and Ruby implementations by OpenAI Codex. They share neither implementation code nor runtime libraries beyond their standard libraries, but they do share the published certificate specification and input format. Thus "independent" means implementation-independent, not independent human authorship or statistically uncorrelated errors. The downstream pair was separately implemented in Python and Ruby by Claude Fable 5. No theorem-critical checker has yet received an independent human line-by-line audit. The small source files, negative controls, and exact replay commands are public so that this remaining software-review task is explicit.

Inputs to Theorem 1 and Proposition 3

Theorem 1 has no mathematical input beyond Lemma 2; its trust rests on the two small checkers. Proposition 3 rests on standard inputs of PL topology: the ribbon-thickening and bistellar-trace interpretations that read the paper's figures into complexes; classification of compact surfaces; local-flatness and normal-block criteria; and simplicial fundamental-group presentations with Tietze theory. The regular-neighborhood and local-flatness conventions are those of [5, Chapters 2–4] and [6, Chapters 3–5]; the presentation moves are the standard ones of [7]. These enter together with the Kerékjártó periodic-disk involution theorem [4] behind the disk extension of the paper's Lemma 7.1, and [3] as corroboration of the framing identification. The ribbon-interpretation input is narrower than the phrase suggests: the certified classification of equivariant ribbon rotations pins the implication from the paper's stated marked five-chain data to the equivariant normal form, and the stated data themselves are audited in the original Lidman–Piccirillo source, so what this boundary retains is the thickening of that data into complexes together with the dashed-arc convention of Section 2. This note reproves none of these.

Inputs to Theorem 4

The twenty-five external theorems of Appendix A, each stated there with its hypotheses. The deep ones are Freedman's classification, the Hambleton–Kreck classification, the invariance theorems for symplectic Kodaira dimension (Li, Ho–Li), the symplectic Thom theorem, Klug's relative Rochlin theorem, and the Floer-theoretic lemmas of Lidman–Piccirillo with the Ozsváth–Szabó nonvanishing theorem behind them; the rest are elementary or are constructions of [2] that this note takes as stated. None is reproved. The repository's dependency ledger records the whole verification as an acyclic graph of 33 named external theorems, 37 machine certificates, 11 geometric arguments, and 20 derived claims ending in the three theorems, with every evidence file bound by hash. These are not competing counts: the 25/3/15/16 figures in Appendix A describe only the compact downstream chain, while 33/37/11/20 describe the larger project-wide ledger that also contains the model, peripheral, framing, and PL verification.

Data availability

This verification targets Wuebben's arXiv:2608.17267v1 (18 August 2026), whose PDF has SHA-256

`16a6cef4998699f76ee508e062f8192cf80eeb478b7e077bfa35ccc25350186a`

The target-version ledger in `TARGET.md` also records the Lidman–Piccirillo v1 digest. All permanent certificates, frozen inputs, replay scripts, hash manifests, run transcripts, and provenance records are contained, with clean-checkout replay instructions, prepared for release as `v1.5.8` of the repository: `https://github.com/VentiMath/exotic-s2xs2-verification/releases/tag/v1.5.8`. This draft must not be circulated until that URL resolves. Once published, the `verification/` tree of that release will be identical to that of the repository commit

`05bda8e474d02cdc0bd0c3ecc29a018cf8a2ffb0`

its filled-group proof manifest has SHA-256

`0cec91193c19b50d496a6d2230d6b6e1dcecb52df60b25117231f31ed21cdae6`

and its downstream-chain certificate has SHA-256

`d564d678b13aec27e9887a508e53c3a1af736436ac3977fe4297297b092daf67`

The repository itself is public at `https://github.com/VentiMath/exotic-s2xs2-verification`; the versioned release link above, rather than search-engine indexing, is the artifact to check. The raw completion histories from which certificates are compiled are reproducible build products and are omitted. The paper's own code is at `https://github.com/bwuebben/exotic-s2xs2`; nothing in the verification derives from it except where explicitly cited.

C Supporting computational history

The proof of Theorem 1 uses only the sealed three-generator chain. An earlier export of the same marked complement had four generators and 95 relators. Its four $n = 0$ certificates contain respectively 1123, 1075, 804, and 1504 retained records and reach the same verdicts, but the interpreter hash seed used to number its raw simplices was not recorded; its input Tietze trace therefore cannot be regenerated. It is retained as corroborating history and no ledger node depends on it.

Four adjacent $n = 1$ presentations replace Ax by $Ar^{-1} = Ax(rx)^{-1}$. They pass the same full-inventory certificate pipeline and check Wuebben's one-step re-indexing, but they are not used for the fixed $n = 0$ member. Likewise, the framing-robustness scan tests 96 cross-and-diagonal meridian shifts plus four zero-shift controls: 22 have derivation certificates and 78 have retained GAP session verdicts. All 100 were reported trivial. Since these counterfactuals do not distinguish the correct framing, the scan is explicitly auxiliary.

The implementation has both positive and negative calibrations. On the standard T^4 Luttinger example it returns the complement $\mathbb{Z}^2 \times F_2$ and the textbook quotient $\mathbb{Z}^4$; on a Baldridge–Kirk example it separates the two sign conventions by trace fingerprints 1 and 3. A 96-relator common core completes to an infinite cyclic group, and a mapping-torus control distinguished $H_1 = \mathbb{Z}^2$ from the $H_1 = \mathbb{Z}^3$ identity case, exposing a construction error that was then fixed. Both filled-group checkers reject a wrong identity root, an altered input relator, an altered rewrite trace, a wrong generator count, an incomplete inventory, and a duplicated certificate batch. These tests show that the pipeline neither collapses every input nor accepts every file; the positive soundness claim remains Lemma 2.

References

[1] B. J. Wuebben, *An exotic $S^2 \times S^2$ and an exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$*, arXiv:2608.17267v1 (2026).

[2] T. Lidman and L. Piccirillo, *Distinguishing closed 4-manifolds by slicing*, arXiv:2505.14387v1 (2025).

[3] D. Auroux, S. K. Donaldson, and L. Katzarkov, *Luttinger surgery along Lagrangian tori and non-isotopy for singular symplectic plane curves*, Math. Ann. **326** (2003), 185–203.

[4] A. Constantin and B. Kolev, *The theorem of Kerékjártó on periodic homeomorphisms of the disc and the sphere*, Enseign. Math. (2) **40** (1994), 193–204.

[5] C. P. Rourke and B. J. Sanderson, *Introduction to Piecewise-Linear Topology*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 69, Springer-Verlag, 1972.

[6] J. F. P. Hudson, *Piecewise Linear Topology*, W. A. Benjamin, 1969.

[7] R. C. Lyndon and P. E. Schupp, *Combinatorial Group Theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 89, Springer-Verlag, 1977.

[8] K. M. Luttinger, *Lagrangian tori in $\mathbb{R}^4$*, J. Differential Geom. **42** (1995), 220–228.

[9] M. H. Freedman, *The topology of four-dimensional manifolds*, J. Differential Geom. **17** (1982), 357–453.

[10] I. Hambleton and M. Kreck, *On the classification of topological 4-manifolds with finite fundamental group*, Math. Ann. **280** (1988), 85–104.

[11] I. Hambleton and M. Kreck, *Cancellation, elliptic surfaces and the topology of certain four-manifolds*, J. Reine Angew. Math. **444** (1993), 79–100.

[12] I. Hambleton and M. Kreck, *On the classification of topological 4-manifolds with finite fundamental group: corrigendum*, Math. Ann. **372** (2018), 527–530.

[13] I. Hambleton, *Intersection forms, fundamental groups and 4-manifolds*, Proceedings of the 15th Gökova Geometry-Topology Conference (2008), 137–150.

[14] C.-I. Ho and T.-J. Li, *Luttinger surgery and Kodaira dimension*, Asian J. Math. **16** (2012), 299–318.

[15] T.-J. Li, *Symplectic 4-manifolds with Kodaira dimension zero*, J. Differential Geom. **74** (2006), 321–352.

[16] W. P. Thurston, *Some simple examples of symplectic manifolds*, Proc. Amer. Math. Soc. **55** (1976), 467–468.

[17] P. Ozsváth and Z. Szabó, *The symplectic Thom conjecture*, Ann. of Math. (2) **151** (2000), no. 1, 93–124.

[18] P. Ozsváth and Z. Szabó, *Holomorphic triangle invariants and the topology of symplectic four-manifolds*, Duke Math. J. **121** (2004), 1–34.

[19] P. Ozsváth and Z. Szabó, *Holomorphic triangles and invariants for smooth four-manifolds*, Adv. Math. **202** (2006), no. 2, 326–400.

[20] A. I. Stipsicz and Z. Szabó, *On the minimal genus problem in four-manifolds*, arXiv:2307.04202 (2023).

[21] M. R. Klug, *A relative version of Rochlin's theorem*, arXiv:2011.12418 (2021).

[22] J. Levine, *Polynomial invariants of knots of codimension two*, Ann. of Math. (2) **84** (1966), 537–554.

[23] A. Kawauchi, *Rational-slice knots via strongly negative-amphicheiral knots*, Commun. Math. Res. **25** (2009), 177–192.

[24] R. S. Palais, *Extending diffeomorphisms*, Proc. Amer. Math. Soc. **11** (1960), 274–277.

[25] D. F. Holt, *KBMAG — Knuth–Bendix in Monoids and Automatic Groups*, software package, version 1.5.9.

[26] The GAP Group, *GAP — Groups, Algorithms, and Programming*, `https://www.gap-system.org`.